INCH-POUND

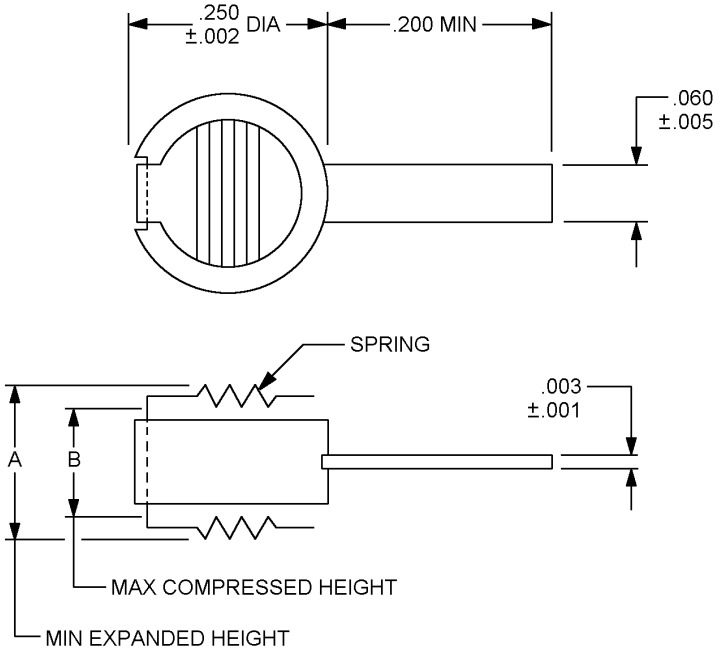
MIL-DTL-39030/21B
02 September 2017
SUPERSEDING
MIL-DTL-39030/21A
14 April 2003

DETAIL SPECIFICATION SHEET

# DUMMY LOADS, ELECTRICAL, TYPE XIV (TAB CONTACT), STRIPLINE, MEDIUM POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-39030.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .060 | 1.52 |
| .121 | 3.07 |
| .140 | 3.56 |
| .200 | 5.08 |
| .246 | 6.25 |
| .250 | 6.35 |
| .265 | 6.73 |

| Dash number | A | B |
|---|---|---|
| 01 | .265 | .246 |
| 02 | .140 | .121 |
| 03 | .265 | .246 |

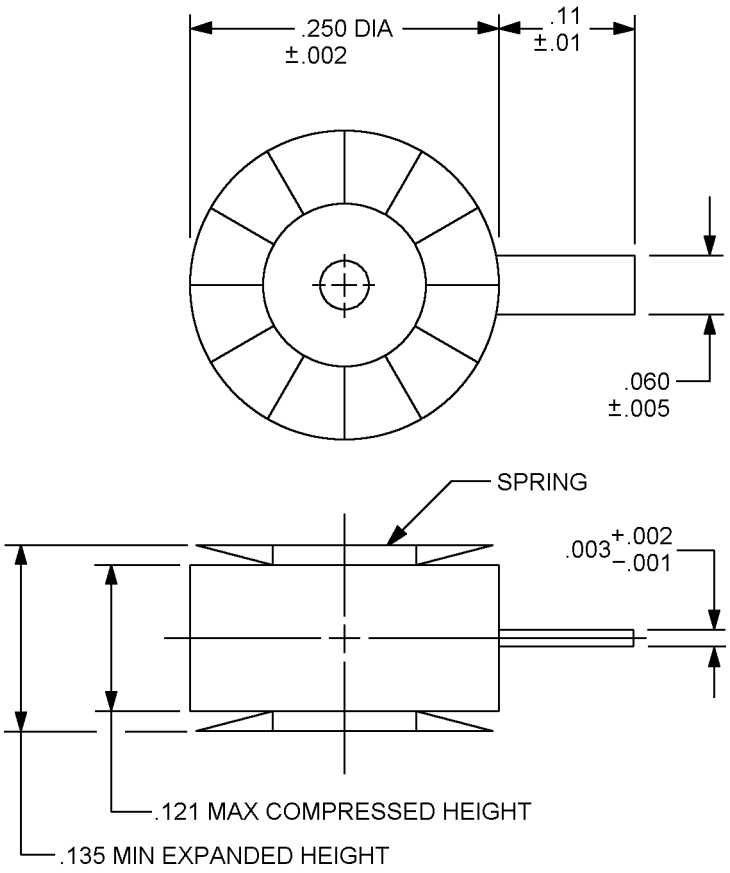
NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Part or Identifying Number (PIN).

FIGURE 1. Dimensions and configuration, PINs M39030/21-01, -02 and -03.

AMSC N/A

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:54Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .01 | .3 |
| .060 | 1.52 |
| .11 | 2.8 |
| .121 | 3.7 |
| .135 | 3.43 |
| .250 | 6.35 |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.

FIGURE 2. Dimensions and configuration, PIN M39030/21-04.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:54Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics.

| Dash number | Operating frequency (GHz) | VSWR (max) | Power handling capability (max) | | Nominal characteristic impedance (ohms) | Weight (max) (oz) | Figure number |
|---|---|---|---|---|---|---|---|
| | | | Average (watts) | Peak (watts) | | | |
| 01 | DC to 3.0 | 1.15:1 1/ | 3 | .1K | 50 | .053 | 1 |
| 02 | DC to 12.0 | (1.20 + .025f):1 2/ | 3 | .1K | 50 | .053 | 1 |
| 03 | DC to 6.0 | (1.10 + .025f):1 2/ | 3 | .1K | 50 | .021 | 1 |
| 04 | DC to 12.4 | 1.10:1 (DC to 10.0 GHz)<br>1.25:1 (10.1 to 12.4 GHz) | 3 3/ | 3K 3/ | 50 | .021 | 2 |

1/ VSWR is specified at 1.8 and 3.0 GHz.
2/ Where “f” is the frequency in GHz.
3/ Input power is derated linearly from 100 at +75°C to 10 percent at +142°C.

REQUIREMENTS:

Dimension and configurations: See figures 1 and 2.

Electrical characteristics: See table I.

Materials:

Body: Brass in accordance with ASTM B121/B121M, ASTM B36/B36M, ASTM B16/B16M, and ASTM B124/B124M, ASTM B21/B21M, ASTM B124/B124M and ASTM B283, QQ-B-639 or ASTM B36 and ASTM B121/B121M.

Finish: Gold plated in accordance with ASTM B488, type 3, grade C, class 1.27 over copper flash.

Tab contact: Brass in accordance with ASTM B121/B121M, ASTM B36/B36M, ASTM B16/B16M, and ASTM B124/B124M, ASTM B21/B21M, ASTM B124/B124M and ASTM B283 or QQ-B-639 (dash numbers 01 thru 03) or beryllium copper in accordance with ASTM B196/B196M, ASTM B197/B197M and ASTM B194 (dash number 04).

Finish: The tab shall be a minimum gold thickness of 50 micro inches (1.27μm) in accordance with ASTM B488, type 3, grade C, class 1.27, over 50 micro inches (1.27 μm) minimum of nickel in accordance with SAE-AMS-QQ-N-290, class 1, measured anywhere along the mating surface, for all series.

Spring: Beryllium copper in accordance with ASTM B196/B196M, ASTM B197/B197M and ASTM B194.

Finish: Gold plated in accordance with ASTM B488, type 3, grade C, class 1.27.

Weight: See table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:54Z
Check the source to verify that this is the current version before use.

Ambient temperature range:

Operating:

Dash numbers 01 thru 04: -55°C to +125°C.
Dash numbers 02 and 03: -65°C to +150°C.

Nonoperating (storage):

Dash numbers 01 and 04: -55°C to +125°C.
Dash numbers 02 and 03: -65°C to +150°C.

Solderability: MIL-STD-202-208, the following details and exception shall apply:

a. Number of terminations of each part to be tested: One contact.

b. Special preparation of terminations: No wiping, cleaning, scraping, or abrasive cleaning of the contact shall be performed.

c. Depth of immersion: The entire surface of the contact shall be covered.

d. Method of mounting: The dummy load shall be mounted onto a heat sink.

e. Examination of terminations: There shall be no evidence of pinholes and blistering.

Note: This test shall be preformed after the visual and mechanical inspection in qualification.

PIN: M39030/21-(dash number from table I).

In addition to MIL-DTL-39030, this document references the following:

| | |
|---|---|
| ASTM B121/B121M | ASTM B283 |
| ASTM B124/B124M | ASTM B36/B36M |
| ASTM B16/B16M | ASTM B488 |
| ASTM B194 | MIL-STD-202-208 |
| ASTM B196/B196M | QQ-B-639C |
| ASTM B197/B197M | SAE-AMS-QQ-N-290 |
| ASTM B21/B21M | |

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue."



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:54Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2017-010)

Review activities:
Army - AV, MI
Navy - AS, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:54Z
Check the source to verify that this is the current version before use.